



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

Name: Veridiano, Lance Jhedrix M.	Date: 9/24/26
Section: IT-2A	Instructor: Ignacio

- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer: This program is a simple Ticket Management System that lets users track incident tickets raised by different automated bots. It comes preloaded with sample data so there are existing records to work with right away. It provides a menu-driven interface for adding, removing, searching, displaying, and counting tickets until the user chooses to exit.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident: `tickets.append({"id": ticket_id, "bot": ticket_bot, "description": ticket_description})`
the information from the input of add appends/ adds from my list

Display Incidents:

For ticket in tickets:

```
print(f'{ticket["id"]}, {ticket["bot"]}, {ticket["description"]}')  
loops of every data in list
```

Search Incident: `if ticket["id"] == ticket_id:`

```
    print (f'INFO FOUND: {ticket["id"]}, {ticket["bot"]},  
    {ticket["description"]}')
```

The information find the entire ID from the list name data until it match and found



Remove Resolved Incident: `if ticket["id"] == ticket_id:`

`tickets.remove(ticket)`

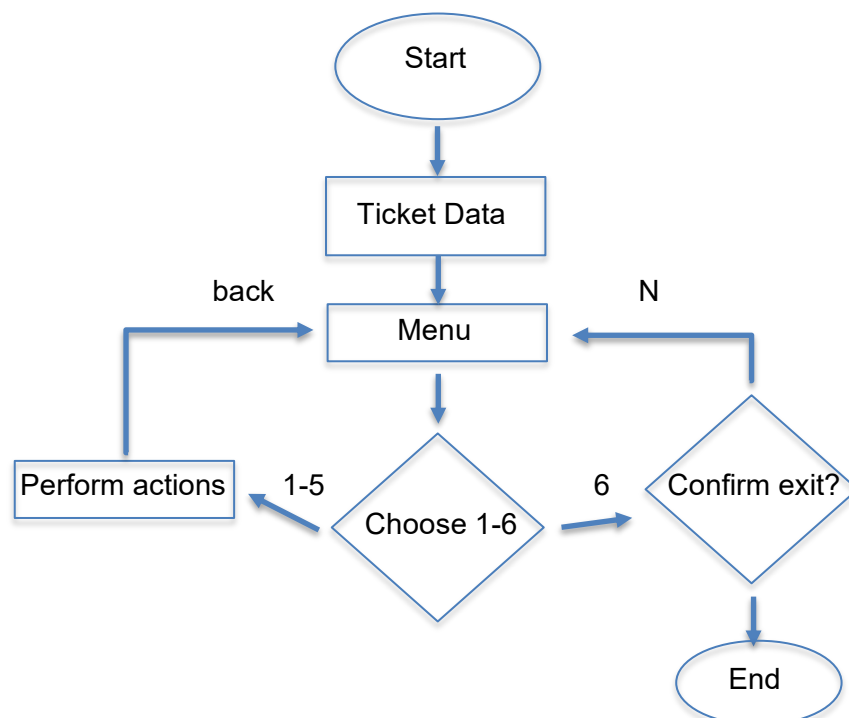
after to type, finding the match value from input then deletes in the list

Count Active Incidents:

`print(f"\nTotal number of tickets: {len(tickets)}")`

printing the total length or numbers of id on list

III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.





Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

- IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

From this activity, I learned how to use a list as a linear data structure to store and manage ticket records, applying basic operations like insertion, searching, deletion, and counting through loops and conditionals. The tickets list demonstrated this directly append () added new elements at the end, for loops traversed the list sequentially to search and remove specific tickets, and len() counted the total elements.
